

Each satellite norms its coefficient preimage in absolute value

MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage

theorem

Varying one functional at an absolute maximizer forces endpoint attainment.

Statement

Let $n \in \mathbb{N}$ and use reference coordinates $V = \mathbb{R}^n$ with the sup norm. Let $p : V \rightarrow \mathbb{R}$ be a continuous seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

Thus p is a norm. We use p for the model distance and $\|\cdot\|_\infty$ for reference-coordinate estimates; the two norms are allowed to coincide. A p -dual contraction is a continuous real linear functional r satisfying $|r(x)| \leq p(x)$ for every x . Let $e_1, \dots, e_n$ be the standard coordinate vectors. For an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals on V , put

$$M_F = (\ell_i(e_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F x, \quad \mathcal{U}_p = \{F : \forall i, x, |\ell_i(x)| \leq p(x)\}.$$

For a real number η , define the attained maximum and near-maximal set by

$$D_p = \max_{F \in \mathcal{U}_p} |\det M_F| > 0, \quad \mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq |\det M_F|\}.$$

Admissibility is measured with p ; the matrices are written in the fixed reference coordinates.

Let J be a finite index set, possibly empty; choose coefficients $c_a \in V$ for $a \in J$. A configuration is a pair $Q = (F, s)$, where F is a frame and $s = (s_a)_{a \in J}$ is a family of additional continuous real linear functionals, called satellites. Write

$$\Omega_p^J = \{(F, s) : F \in \mathcal{U}_p, \forall a, x, |s_a(x)| \leq p(x)\}.$$

For any real weight w , let $M_F[i \leftarrow r]$ replace row i by $(r(e_j))_j$ and set

$$\Phi_{w,c}(F, s) = w \det M_F + \sum_{a \in J} \sum_{i=1}^n (c_a)_i \det M_F[i \leftarrow s_a].$$

An absolute maximizer means a single $Q \in \Omega_p^J$ with $|\Phi_{w,c}(Q')| \leq |\Phi_{w,c}(Q)|$ for every $Q' \in \Omega_p^J$. It need not maximize the signed polynomial. Suppose $\eta < D_p$ and the admissible absolute maximizer $Q = (F, s)$ also has $F \in \mathcal{N}_{p,\eta}$. For every fixed $a \in J$, put $z_a = M_F^{-1} c_a$. Then

$$|s_a(z_a)| = p(z_a).$$

Assumptions

Finiteness of J , the positive comparison constants for p , the positive determinant gap, admissibility, near-maximality of the base, and absolute maximality are required. The weight is any real number. No positive-weight or budget assumption is added.

Conclusion

Every satellite attains the norm in absolute value at its own coefficient preimage. This gives neither $s_a(z_a) = p(z_a)$ nor a common positive sign for all satellites.

Notes

The configuration polynomial evaluates Φ on a pair. the product configuration type, satellite families, their contraction subset and the admissible product set retain both components of Q . The selected absolute maximizer is a choice from the compact product, not a uniqueness assertion. The proof-local scalar called p in the source is the number $p(z_a)$ here; it is not a second norm.

Proof route

Separate one satellite contribution, use dominated Hahn–Banach endpoints, and apply rigidity of an affine absolute maximum.

▼ Proof steps

1. **Keep the coefficient preimage visible.** Fix $a \in J$ and keep $z_a = M_F^{-1}c_a$ as in the statement. The gap gives $\det M_F \neq 0$. Collect the terms that do not involve the a th satellite in

$$R_a = w \det M_F + \sum_{b \neq a} \sum_i (c_b)_i \det M_F[i \leftarrow s_b].$$

For every replacement functional r , row Cramer for this same matrix and coefficient column gives

$$\sum_i (c_a)_i \det M_F[i \leftarrow r] = (\det M_F) r(M_F^{-1}c_a) = (\det M_F) r(z_a).$$

Substituting just this row contribution into the polynomial gives

$$\begin{aligned} \Phi_{w,c}(F, s[a \leftarrow r]) &= R_a + (\det M_F) r(z_a), \\ \Phi_{w,c}(F, s) &= R_a + (\det M_F) s_a(z_a). \end{aligned}$$

This is the one-satellite update formula, with the same F , c_a and nonzero determinant. Only the functional in position a changes; R_a and z_a stay fixed. Admissibility gives $|s_a(z_a)| \leq p(z_a)$.

2. **Supply the two norming endpoints at that same point.** The required functionals satisfy

$$|r_{\pm}(x)| \leq p(x) \quad (x \in V), \quad r_+(z_a) = p(z_a), \quad r_-(z_a) = -p(z_a).$$

If $z_a = 0$, take both functionals to be zero. Otherwise define h on the line $\mathbb{R}z_a$ by $h(\beta z_a) = \beta p(z_a)$. Its domination is

$$h(\beta z_a) = \beta p(z_a) \leq |\beta| p(z_a) = p(\beta z_a).$$

Apply continuous seminorm-dominated Hahn–Banach extension with subspace $\mathbb{R}z_a$, this h , and continuous seminorm p . It gives a continuous linear extension r_+ with $r_+(x) \leq p(x)$ for all x . Applying this inequality to $-x$ also gives $|r_+(x)| \leq p(x)$. Moreover $r_+(z_a) = h(z_a) = p(z_a)$. Set $r_- = -r_+$. These are the functionals in the norming endpoint lemma. Replacing only s_a with either of them preserves every contraction condition, so both new configurations belong to Ω_p^J .

3. **Compare the actual value with both endpoints.** Absolute maximality, applied to the two configurations from Step 2 and then to the formula in Step 1, gives

$$\begin{aligned} |R_a + (\det M_F) p(z_a)| &\leq |R_a + (\det M_F) s_a(z_a)|, \\ |R_a - (\det M_F) p(z_a)| &\leq |R_a + (\det M_F) s_a(z_a)|. \end{aligned}$$

Since $p(z_a) \geq 0$, the endpoint identity is

$$\max\{|R_a + (\det M_F) p(z_a)|, |R_a - (\det M_F) p(z_a)|\} = |R_a| + |\det M_F| p(z_a).$$

Combining it with the two comparisons and the triangle inequality yields

$$\begin{aligned} |R_a| + |\det M_F| p(z_a) &\leq |R_a + (\det M_F) s_a(z_a)| \\ &\leq |R_a| + |\det M_F| |s_a(z_a)| \\ &\leq |R_a| + |\det M_F| p(z_a). \end{aligned}$$

All inequalities are equalities. Cancel $|R_a|$ and divide by $|\det M_F| > 0$ to obtain $|s_a(z_a)| = p(z_a)$. This is absolute affine endpoint rigidity applied directly to the nonnegative number $p(z_a)$, the value $s_a(z_a)$, and these two endpoint comparisons. The case $p(z_a) = 0$ is included.

Main citations

- one-satellite update formula — `MathlibAnnex.Satellite.Internal.configurationPolynomial_update_eq`
- continuous seminorm-dominated Hahn–Banach extension — `Module.Dual.exists_continuous_extension_of_le_seminorm_real`
- the norming endpoint lemma — `MathlibAnnex.Satellite.Internal.exists_modelNormingEndpoints`
- absolute affine endpoint rigidity — `MathlibAnnex.Satellite.abs_affine_endpoint_rigidity`
- configuration polynomial — `MathlibAnnex.Satellite.configurationPolynomial`
- the product configuration type — `MathlibAnnex.Satellite.SatelliteConfiguration`
- satellite families — `MathlibAnnex.Satellite.SatelliteRows`
- their contraction subset — `MathlibAnnex.Satellite.satelliteRowsSet`
- the admissible product set — `MathlibAnnex.Satellite.satelliteConfigurationSet`
- selected absolute maximizer — `MathlibAnnex.Satellite.maxSatelliteConfiguration`
- Exact declaration and proof — `MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage`

Lean source signature (exact)

```
theorem absoluteMaximizer_satellite_norms_preimage {n : ℕ}
  (M : NormModel n) {J : Type u} [Fintype J] [DecidableEq J]
  {η weight : ℝ} (hηD : η < detMax M)
  (coeff : J → Coord n)
  {C : SatelliteConfiguration n J}
  (hC : C ∈ satelliteConfigurationSet M J)
  (hnear : C.1 ∈ nearMaxFrames M η)
  (hmax : ∀ D ∈ satelliteConfigurationSet M J,
    |configurationPolynomial weight coeff D| ≤
    |configurationPolynomial weight coeff C|)
  (a : J) :
  |C.2 a (framePreimage C.1 (coeff a))| =
  M.p (framePreimage C.1 (coeff a))
```

The index a is arbitrary but supplied to the theorem. The conclusion concerns its already fixed satellite; it does not choose a new functional.

In the source	Mathematical meaning
<code>Coord n</code>	The reference space $V = \mathbb{R}^n$ with its sup norm; <code>Coord n</code> abbreviates $\text{Fin } n \rightarrow \mathbb{R}$. Lean uses indices $0, \dots, n-1$ and the formulas use $1, \dots, n$.
<code>M : NormModel n</code> ; <code>M.p</code>	The input M contains the continuous seminorm p and constants $a_p, b_p > 0$ with $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every x . <code>M.p</code> x is the scalar $p(x)$.
<code>[Fintype J]</code> <code>[DecidableEq J]</code>	J is finite, with equality of indices available for replacing one entry. This adds no mathematical restriction to a finite set.
<code>DualFrame n</code> ; <code>dualFrameMatrix B</code> ; <code>frameDet B</code>	An ordered family $F = (\ell_i)$ of continuous real linear functionals, its matrix $M_F = (\ell_i(e_j))_{i,j}$ in the standard basis, and its determinant $\det M_F$. No independence is imposed by the family type.
<code>detMax M</code> ; <code>nearMaxFrames M η</code>	$D_p = \max_{F \in \mathcal{U}_p} \det M_F $ and $\mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq \det M_F \}$, where $\mathcal{U}_p = \{(\ell_i) : \forall i, x, \ell_i(x) \leq p(x)\}$.
<code>SatelliteConfiguration n J</code> ; <code>C.1</code> ; <code>C.2 a</code>	A pair $Q = (F, s)$, called C . Its first component <code>C.1</code> is the base frame F ; <code>C.2 a</code> is the satellite $s_a : V \rightarrow \mathbb{R}$, continuous and real linear.
<code>weight</code> ; <code>coeff a</code> ; <code>coeff a i</code>	The real weight w , the coefficient vector $c_a \in V$, and its i th coordinate $(c_a)_i$. These coefficients are fixed before maximization.

In the source	Mathematical meaning
<code>hC : C ∈ satelliteConfigurationSet M J</code>	The given configuration lies in Ω_p^J : $ \ell_i(x) \leq p(x)$ and $ s_a(x) \leq p(x)$ for every i, a, x .
<code>configurationPolynomial weight coeff C</code>	The scalar $\Phi_{w,c}(Q) = w \det M_F + \sum_{a \in J} \sum_i (c_a)_i \det M_F[i \leftarrow s_a]$, where the replaced row is $(s_a(e_j))_j$.
<code>hmax : ∀ D ∈ satelliteConfigurationSet M J, ...</code>	For every admissible comparison pair $Q' \in \Omega_p^J$, $ \Phi_{w,c}(Q') \leq \Phi_{w,c}(Q) $. D names that pair, not the number D_p . This is a hypothesis about absolute values.
<code>hηD; hnear</code>	The hypotheses $\eta < D_p$ and $F \in \mathcal{N}_{p,\eta}$, giving a nonzero determinant for the actual base frame.

In the source	Mathematical meaning
<code>(a : J); framePreimage C.1 (coeff a)</code>	The supplied index a and the vector $z_a = M_F^{-1}c_a$.
<code> C.2 a (framePreimage C.1 (coeff a)) = M.p (framePreimage C.1 (coeff a))</code>	The whole conclusion $ s_a(M_F^{-1}c_a) = p(M_F^{-1}c_a)$. The same coefficient and satellite index are used on both sides.

Here $\rightarrow L[\mathbb{R}]$ means continuous real linear, $\forall$ means “for every”, and $\in$ is membership. Dot notation selects a field or component; successive arguments denote function application. No positivity of weight is assumed here.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 13–15:

```
private abbrev Coord (n : N) := Fin n → ℝ
private abbrev NormModel (n : N) := EquivalentSeminorm (Coord n)
private abbrev DualFrame (n : N) := DeterminantFrame.Frame n (Coord n)
```

Exact source lines 20–21:

```
private abbrev framePreimage {n : N} (B : DualFrame n) (c : Coord n) := (dualFrameMatrix B)-1.mulVec c
private abbrev IsDualContraction {n : N} (M : NormModel n) (r : Coord n → L[ℝ] ℝ) := ∀ x, |r x| ≤ M.p x
```

Exact source lines 53–55:

```
private abbrev detMax {n : N} (M : NormModel n) := DeterminantFrame.determinantMaximum (modelBasis M)
private abbrev NearMaxInverseBound {n : N} (M : NormModel n) := DeterminantFrame.NearMaxInverseBound (modelBasis M)
private abbrev nearMaxFrames {n : N} (M : NormModel n) (η : ℝ) := {B : DualFrame n | B ∈ dualFrameSet M ∧ detMax M - η ≤ |frameDet B|}
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 7–55. The full original context is retained with the source evidence.

▼ **Exact content identity**

Declaration: MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage

Accepted content SHA-256: 97c725784ef9819be6916f9555eb8a2a96132f1aa09f946f8f0359d30f1cd2b1

Accepted source guide SHA-256: 07d5727b5ae19541dbdf756b0d0ef1659410842cfd1422a16f484b84f88e046a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73